

Codebase Map — Visual Guide

One node per modules/<domain>, wired to the wiki concept it implements. The code-side companion to **TVI-CAMS Knowledge Base**.

This repository *is* an Obsidian vault. Two map-of-content hubs sit at its root and answer two different questions:

TVI-CAMS Knowledge Base

Maps the docs/ & wiki/ — *what* the product is and *why*. Your existing guide.

Codebase Map (this guide)

Maps modules/<domain> — *where* each feature lives in code, and how it links back to a concept.

How to not get lost: start at a *concept* you know (right side, e.g. Billing Engine) → follow the link to its *module* (left side, e.g. Billing Module) → open that module's note for the source path and status. In Obsidian's **graph view** the same structure appears as a code cluster fused to the concept cluster. Because the vault hides unresolved links, every node below is a real file.

How to read the map

Node status — the colour of each module box and its link line:

 Implemented  Partial  Stub  Leaf

→ **Left column** = code modules (the graph's codebase cluster).

→ **A link line** = "this module implements this concept."

→ **Right column** = wiki concept notes (the "why").

→ **Line colour** = the module's implementation status.

The map

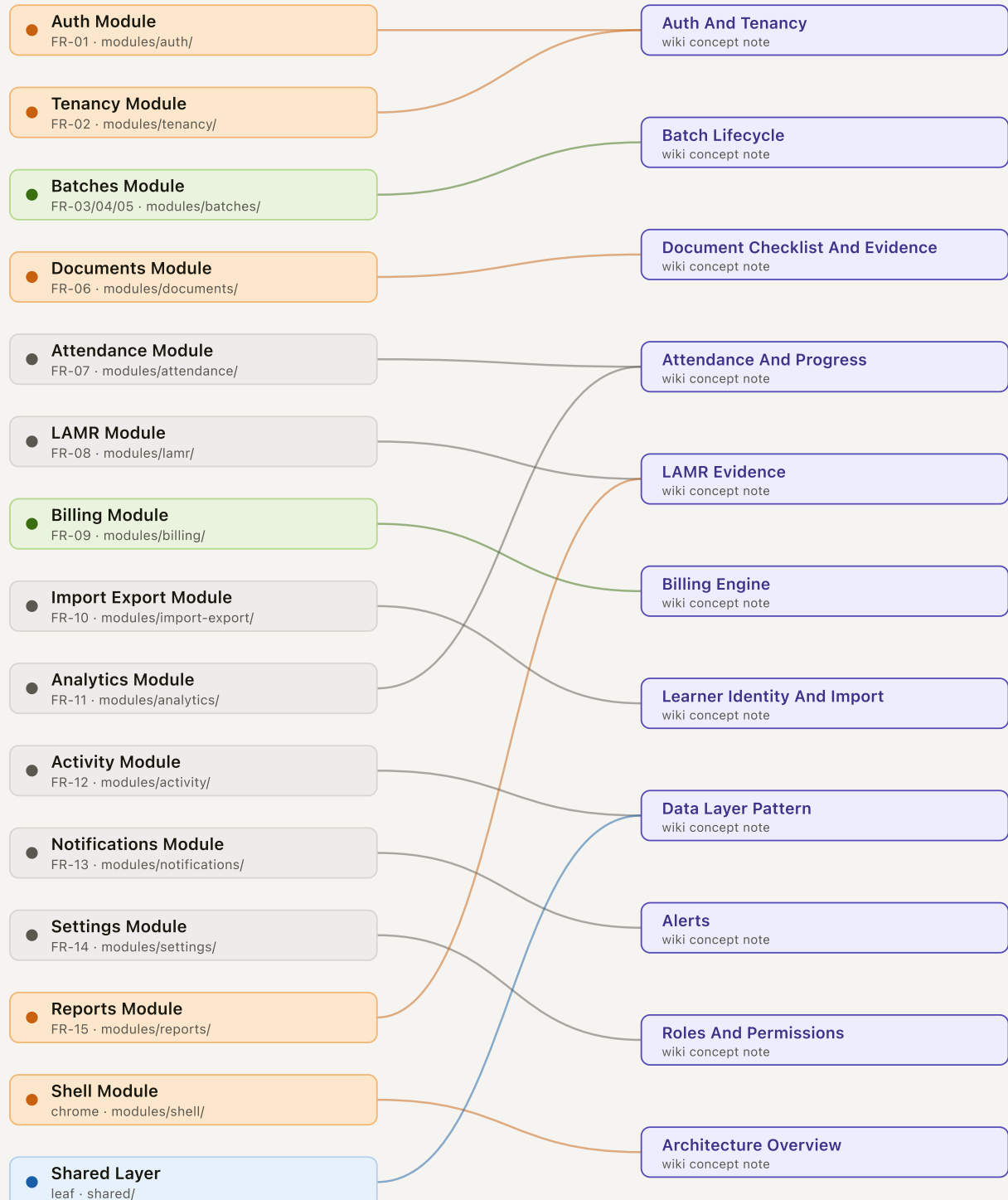
Code modules (left) linked to the wiki concepts they implement (right). Status by colour; see legend on page 1.

CODE — modules/<domain>

each box is a real graph node

CONCEPTS — wiki/

what each module implements



Guided index — modules by functional requirement

The same nodes as a table. Open codebase/<Name>.md for each, or jump to its concept note.

MODULE	FR	STATUS	CONCEPT NOTE	SOURCE PATH
Auth Module Clerk identity + session resolution	FR-01	Partial	Auth And Tenancy	modules/auth/
Tenancy Module Tenant-in-URL context; RLS scopes rows	FR-02	Partial	Auth And Tenancy	modules/tenancy/
Batches Module Dashboard, lifecycle, urgency engine — reference impl	FR-03/04/05	Implemented	Batch Lifecycle	modules/batches/
Documents Module Required docs, verification, evidence store	FR-06	Partial	Document Checklist And Evidence	modules/documents/
Attendance Module Attendance-derived progress; 5-absence rule	FR-07	Stub	Attendance And Progress	modules/attendance/
LAMR Module Learners Achievement Monitoring Report	FR-08	Stub	LAMR Evidence	modules/lamr/
Billing Module Document-generating engine + statement preview	FR-09	Implemented	Billing Engine	modules/billing/
Import Export Module Learner import wizard; ULI as key	FR-10	Stub	Learner Identity And Import	modules/import-export/
Analytics Module Aggregate metrics across batches	FR-11	Stub	Attendance And Progress	modules/analytics/
Activity Module activity_log audit trail	FR-12	Stub	Data Layer Pattern	modules/activity/
Notifications Module In-app alerts, computed on read	FR-13	Stub	Alerts	modules/notifications/
Settings Module Tenant + user settings; role-gated	FR-14	Stub	Roles And Permissions	modules/settings/
Reports Module Cross-batch reports	FR-15	Partial	LAMR Evidence	modules/reports/
Shell Module App chrome: Sidebar, Topbar, MetricsRow	chrome	Partial	Architecture Overview	modules/shell/
Shared Layer Props-only ui, domain types, mock seed	leaf	Leaf	Data Layer Pattern	shared/

How the layers connect

Code is grouped by **domain, not file type** (TES-68). Import direction is ESLint-enforced:

app/ → modules/<domain>/{data, domain, ui} → shared/ → lib/supabase

- app/ — thin routes; Server Components fetch + compose.
- data/ — fetch → map → derive; only layer that may import `database.types`.
- domain/ — pure business rules, no I/O; unit-tested with fixed as-of dates.
- ui/ — domain-aware components; reuse `shared/ui` primitives.

Key cross-cutting facts

- RLS is the security boundary; UI hiding is usability only.
- Trainer DTOs omit billing / NTP-lag / BSRS fields *server-side*.
- Billing generates official `.docx` documents (ADR-001), not a passive signal.
- Enum bridge lives in the mapper, never in components.
- Alerts computed on read — no cron, no email.
- One RQM code = one batch; ULI is the permanent learner key.

Generated 2026-07-14 from Codebase `Map.md + codebase/*.md` and the live `modules/` tree. Colours and type follow the TVI-CAMS design tokens (`app/globals.css`). Statuses reflect code present as of this date (implemented / partial / stub); the wiki concept notes were refreshed the same day for the TES-68 DDD reorg and TES-70 billing module.